MATH290

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Proposition 1. *For any integer k, $k^3 - k$ is a multiple of 6.*

Proof.

Firstly, note that the sum of two multiples of 6 is itself a multiple of 6. An integer is a multiple of 6 if it can be expressed as the product of 6 and some other integer. Consider a multiple of six $n = 6i$ and another $m = 6j$. Observe,

$$\begin{aligned} n + m &= 6i + 6j \\ &= 6(i + j) \end{aligned}$$

Relying on the property that summing integers yields an integer, we've shown that $n + m$ can be expressed as a product of 6 and some other integer $i + j$. That is, $n + m$ is a multiple of 6 for any multiples of six n and m.

Let f_k be the sequence $k^3 - k$ for integers $k \geq 0$. Let g_k be the sequence formed by the *difference* between f_n and f_{n+1} for any $n \in \mathbb{Z}$. That is,

$$f_{k+1} = f_k + g_k$$

□